Yuxuan Song

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Education

Shanghai Jiao Tong University **Shanghai, China**
Sep. 2013-Mar. 2020(expected)

- Bachelor and Master of Science in Computer Science and Technology
- Undergraduate Cumulative GPA: Overall: 87.6/100 (36/138) | Major: 90.2/100
- Research Assistant, Apex Data and Knowledge Management Lab

Research Interests

My research interest lies in representation learning and generative modeling(VAE and GANs). I am also interested in using information-theoretic approaches to improve upon existing deep generative models, and in its application to density estimation, compression and reinforcement learning.

Experiences

Research Intern at Bytedance(TikTok) AI lab **Shanghai,China**
Worked on neural text generation advised by Dr. Hao Zhou and Dr. Lei Li. *June. 2019- Sep. 2019*

- Proposed to improve Maximium-Likelihood-Estimation(MLE) training for text generation by likelihood-free importance sampling, where the likelihood ratio is estimated through **Bregmann divergence** minimization.
- Advanced all previous methods in both quality and diversity. .

Student Researcher at Apex Data and Knowledge Management Lab **Shanghai,China**
Worked on deep learning and data mining advised by Weinan Zhang and Yong Yu. *Aug. 2016-May. 2019*

- Here is a brief introduction to some selected works (in proceedings or in submissions).
  - **Understanding and Analysis of the Unstability in GANs.**
    - Investigate the underlying factor that leads to the failure and success in training of GANs.
    - Provide theoretical analysis and proof on the effect of the **Lipschitz** constraint in GAN.
    - Find and propose a new family of objective functions, and conduct experiments to prove the superiority.
  - **Unsupervised Domain Adaption with Variational Information Bottleneck**
    - Developed a novel approach to unsupervised domain adaptation by leveraging the combination of cluster assumption and **information bottleneck** principle.
    - Employ the information bottleneck module in deep arichitecture with variational approximation.
    - Demonstrated that the model achieves state-of-the-art on several visual domain adaptation benchmarks.
  - **Lossless and Lossy Compression with Latent Variable Model**
    - Analyzed the key limitation of turning VAE as a practical compression algorithm for sequential data.
    - Proposed a new coding scheme for the sequential latent variable models based on **bits-back** coding.
    - Proposed a solution for joint source channel coding by variational mutual information maximization.
    - Advanced the previous work on coding with latent variable model by a lower compression rate.
  - **Top Optimization and Sequential Selection in Computational Advertising**
    - Developed a new ranking algorithm to focus on the top ranked items by strategic sampling methods.
    - Designed a reinforcement learning model for the sequential selection scenario in programmatic advertising.
    - Elevated baselines through 7-day A/B test and acquired more stable click numbers and profits.
    - The ranking and selection system was deployed in a commercial online advertising platform.

Publications

- **Y. Song**, L. Yu, Z. Cao, Z. Zhou, J. Shen, S. Shao, W. Zhang and Y. Yu. Improving Domain Adaptation

with Variational Information Bottleneck. 24th European Conference on Artificial Intelligence, (**ECAI**, to appear), 2020, Arxiv version

- **Y. Song**, N. Miao, H. Zhou, L. Yu and L. Li. Improving Maximum Likelihood Training for Text Generation with Density Ratio Estimation. 23rd International Conference on Artificial Intelligence and Statistics (**AISTATS**, to appear), 2020.
- **Y. Song**, M. Xu, L. Yu, H. Zhou, S. Shao and Y. Yu. Infomax Neural Joint Source-Channel Coding via Adversarial Bit Flip. 34th AAAI Conference on Artificial Intelligence (**AAAI**, to appear), 2020.
- **Y. Song**, H. Cai, K. Ren, W. Zhang and Y. Yu. Volume Ranking and Sequential Selection in Programmatic Display Advertising. In Proceedings of the 26th ACM International Conference on Information and Knowledge Management, pages 1099-1107 (**CIKM**), 2017
- Z. Zhou, J. Liang, **Y. Song**, L. Yu, H. Wang, Z. Zhang, W. Zhang and Y. Yu. Lipschitz Generative Adversarial Nets. In Proceedings of 36th International Conference on Machine Learning (**ICML**), 2019
- G. Lu, Z. Zhou, **Y. Song**, K. Ren and Y. Yu.Guiding the One-to-one Mapping in CycleGAN via Optimal Transport. In Proceedings of the 33rd AAAI Conference on Artificial Intelligence (**AAAI**), 2019
- Z. Zhou, H. Cai, S. Rong, **Y. Song**, K. Ren, W. Zhang, Y. Yu and J. Wang. Activation Maximization Generative Adversarial Nets. In Proceedings of the 6th International Conference on Learning Representations (**ICLR**), 2018

Manuscripts

- Z. Zhou, **Y. Song**, J. Liang, L. Yu, H. Wang, Z. Zhang, W. Zhang and Y. Yu. Understanding the Effectiveness of Lipschitz Constraint in Training GANs via Gradient Analysis. Technical report. arXiv preprint arXiv:1807.00751, 2018.

Honors and Awards

- **Huawei Scholarship**. 2018
- **CIKM'17/SIGIR Student Travel Grant**. 2017
- **Shanghai Jiao Tong University Outstanding Graduate**. 2017
- **Shanghai Jiao Tong University Excellent Scholarship**. 2014&2015&2016
- **The First Prize in National Mathematical Olympiad in Senior**. 2012

Skill Set

- **Machine Learning API:** Tensorflow, Pytorch
- **English Proficiency:** TOEFL 105(R30, L25, S23, W27), GRE 325(V156, Q169, W3.5)